Metadata: StormSense Comma Separated Values (CSV) for 72-Hour Forecasts (fore72)

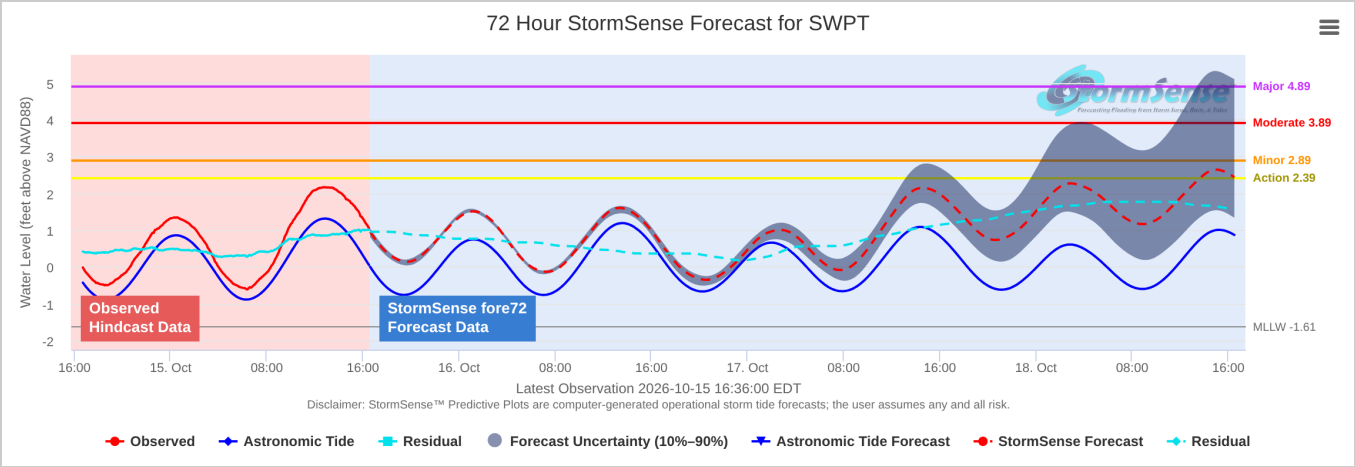


Figure 1. *Example StormSense Chart of 72-hour forecast data (NAVD) at NOAA's Sewells Point tide gauge, in Norfolk, VA, issued October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z) from the P-ETSS 12Z cycle.*

Fields **fore72 data downloads only include data series for the 72-hour forecast period (dashed lines and shaded cone)**

date_time — Date and time in Zulu/GMT, formatted as a string leading with the date as "YYYY-MM-DD" followed by the time (T) as "hh:mm:ss" relative to Zulu (Z) time. Recorded in regular 6-minute intervals (0, 6, 12 … 54 minutes past the hour), listed in chronological order.

astro_tide — (**Astronomic Tide Forecast**) Harmonically predicted astronomical tide for the next 72 hours from the NWS P-ETSS model. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to the North American Vertical Datum of 1988 (NAVD88).

surge — Ensemble-mean storm surge from P-ETSS (the Probabilistic Extra-Tropical Storm Surge model of the NWS Meteorological Development Laboratory, run with an ensemble of weather forecasts): the water level change driven by forecast wind and air pressure. Hourly guidance interpolated to 6 minutes and smoothed with a 3-hour low-pass filter. Feet; a height difference, so the same in MLLW and NAVD88.

bias — The observation-based correction P-ETSS adds to its surge, from the recent difference between the gauge's observations and the modeled tide plus surge; it is held or decays through the forecast, and is 0.000 where P-ETSS applies no correction. Feet; the same in MLLW and NAVD88.

residual — (**Residual**) Forecast residual: the non-tidal part of the forecast, SSforecast − astro_tide (surge + bias). Feet; the same in MLLW and NAVD88.

SSforecast — (**StormSense Forecast**) The StormSense 72-hour storm tide forecast, the ensemble-mean total water level: astro_tide + surge + bias. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to the North American Vertical Datum of 1988 (NAVD88).

SSforecast_p10 — (**Forecast Uncertainty (10%–90%)**) 10th percentile of the ensemble total water level (P-ETSS's 90% exceedance level): the lower edge of the shaded cone of uncertainty. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to the North American Vertical Datum of 1988 (NAVD88).

SSforecast_p90 Upper edge of the cone: the 90th percentile of the ensemble total water level (P-ETSS's 10% exceedance level). Nine in ten ensemble members fall at or below it. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to the North American Vertical Datum of 1988 (NAVD88).

surge_p10, surge_p90 10th and 90th percentiles of the ensemble storm surge (P-ETSS 90% and 10% exceedance). Feet; the same in MLLW and NAVD88.

P-ETSS levels are tied to each gauge's own record: the median difference between the observations and P-ETSS over the 24 hours before the forecast starts is added to astro_tide, SSforecast and its percentiles (0.00 ft at most NOAA gauges). At stations without a P-ETSS point, the file keeps the same columns with the 36-hour forecast in astro_tide, residual and SSforecast and the P-ETSS-only columns left blank.

Table 1. *Example StormSense Table of 72-hour forecast data (NAVD) at NOAA's Sewells Point tide gauge, in Norfolk, VA, issued October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z).*

date_time	astro_tide	surge	bias	residual	SSforecast	SSforecast_p10	SSforecast_p90	surge_p10	surge_p90
2026-10-15T20:36:00Z	0.000	1.000	-0.052	0.948	0.947	0.807	1.047	0.860	1.100
2026-10-15T20:42:00Z	-0.050	1.000	-0.052	0.948	0.897	0.762	0.997	0.864	1.100
2026-10-15T20:48:00Z	-0.099	1.000	-0.052	0.948	0.849	0.718	0.949	0.869	1.100
2026-10-15T20:54:00Z	-0.145	1.000	-0.052	0.948	0.803	0.676	0.902	0.873	1.100
2026-10-15T21:00:00Z	-0.192	1.000	-0.052	0.948	0.757	0.634	0.856	0.877	1.099
2026-10-15T21:06:00Z	-0.236	1.000	-0.052	0.948	0.712	0.593	0.811	0.882	1.099
2026-10-15T21:12:00Z	-0.279	1.000	-0.051	0.949	0.669	0.554	0.767	0.885	1.098
2026-10-15T21:18:00Z	-0.321	1.000	-0.051	0.949	0.627	0.516	0.724	0.889	1.097
2026-10-15T21:24:00Z	-0.361	1.000	-0.051	0.949	0.587	0.479	0.683	0.891	1.096
2026-10-15T21:30:00Z	-0.400	1.000	-0.051	0.949	0.548	0.442	0.642	0.894	1.094
2026-10-15T21:36:00Z	-0.436	1.000	-0.051	0.949	0.513	0.408	0.604	0.896	1.091
2026-10-15T21:42:00Z	-0.472	1.000	-0.051	0.949	0.478	0.375	0.566	0.897	1.089

NWS Flood Thresholds *(colored lines labeled at the right of the chart; click a label to hide or show its line)*

The flood categories on NAVD88 charts are defined by the National Weather Service (NWS), not by StormSense. NWS forecast offices set them for each forecast gauge together with local emergency managers, based on the impacts observed near that gauge, and publish them through the National Water Prediction Service. StormSense refreshes them from NWPS with every update and shows them only at the stations tied to an NWS gauge; thresholds that NWS publishes in MLLW are shifted to NAVD88 with the station's MLLW–NAVD88 separation.

Action (yellow) The level at which the NWS and its partners begin preparing for possible flooding; water may reach the edges of low-lying areas, but significant impacts are not yet expected.

Minor (orange) Minor flooding: minimal or no property damage, but possibly some public threat, such as water over low-lying roads, parking lots and parks near the shore.

Moderate (red) Moderate flooding: some inundation of structures and roads near the water, with some evacuations of people or moving of property to higher ground.

Major (purple) Major flooding: extensive inundation of structures and roads, with significant evacuations of people and transfer of property to higher elevations.

MLLW translation datum

—— **MLLW** Mean Lower Low Water: the average of the lower of each day's two low tides over the National Tidal Datum Epoch (1983–2001). The gray MLLW line marks where the MLLW zero (the chart datum used on nautical charts and in tide tables) sits on the NAVD88 scale, labeled with its NAVD88 elevation.

Datum source: for most NOAA-operated tide gauges the MLLW–NAVD88 separation is NOAA's published value for that station. USGS, VDEM and StormSense sensors natively report live water levels in NAVD88 only, so their MLLW translation datum is derived from NOAA VDatum (https://vdatum.noaa.gov/vdatumweb/).